ALGORITHM

1. Read the number of offices.
2. Read the cost matrix representing the fiber-optic connections.
3. Replace zero values between different offices with infinity to indicate the absence of a direct link.
4. Ask the user to enter the two offices whose edge weight must be updated.
5. Read the new edge weight and update the cost matrix in both directions.
6. Initialize all offices as unvisited and select the first office as the starting vertex.
7. Find the minimum-weight edge connecting a visited office to an unvisited office.
8. If no valid edge exists, report that the graph is disconnected and a spanning tree cannot be formed.
9. Otherwise, select the edge and display its weight.
10. Add the edge weight to the running total and display the updated total.
11. Mark the newly connected office as visited.
12. Repeat Steps 8–12 until all offices are connected.
13. Display the final minimum cable cost.
14. Stop.